

SECTION 11: CHILLER SYSTEMS SOP

Purpose: Ensure safe, reliable, and efficient operation of centrifugal, screw, or absorption chillers. These procedures reduce mechanical failure, maintain cooling performance, and ensure chemical and refrigerant compliance.

Daily Operational Procedures:

Conduct visual inspection of chiller and surrounding area for leaks or abnormal sounds.

Record chilled water supply/return temperatures.

Verify condenser water inlet/outlet temperatures.

Check oil level, pressure, and refrigerant levels.

Ensure all interlocks, safeties, and sensors are online.

Inspect strainers and differential pressure across heat exchangers.

Monitor vibration and log compressor run hours.

Weekly Maintenance Tasks:

Clean or backflush strainers and filters.

Verify pressure/temperature transducer accuracy.

Check insulation on piping for damage or sweating.

Record current draw on compressor motors.

Safety Notes:

Always wear eye protection when inspecting refrigerant sight glasses.

Follow all EPA guidelines for refrigerant handling.

Applicable Codes & Guidelines:

ASHRAE 15 & 34

EPA Section 608

NFPA 70 (electrical safety)